

BOONE COUNTY AP, AR, USA

WMO: 723450

Lat: 36.267N

Lon: 93.157W

Elev: 419

StdP: 96.39

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13971

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-8.8	-17.7	0.8	-8.9	-15.3	1.0	-6.2	10.5	1.2	9.3	4.9	3.5	310	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	35.1	23.4	33.4	23.5	32.2	23.2	25.3	31.4	24.7	30.7	24.1	29.9	3.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.4	28.4	22.9	18.6	27.6	22.4	18.0	27.1	79.8	31.5	76.9	30.6	74.6	30.0	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	8.0	7.2	DB	-15.2	36.8	2.1	2.4	-16.7	38.5	-17.9	39.9	-19.0	41.3	-20.5	43.1	(4)
(5)			WB	-15.7	26.7	2.0	1.4	-17.1	27.7	-18.3	28.6	-19.4	29.4	-20.8	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	3.0	5.0	9.5	14.9	19.2	23.7	25.9	25.5	21.3	15.5	9.5	4.3
	DBStd	9.40	6.54	6.36	5.91	4.76	4.10	2.91	2.75	2.97	3.82	4.74	5.34	5.71
	HDD10.0	764	232	163	81	13	1	0	0	0	0	9	73	191
	HDD18.3	2122	476	375	277	121	40	2	0	1	16	113	268	434
	CDD10.0	2525	15	22	67	161	286	411	492	480	339	179	58	16
	CDD18.3	841	0	1	4	18	67	163	234	222	104	25	2	0
	CDH23.3	7841	0	3	36	147	497	1479	2369	2188	913	194	14	0
	CDH26.7	3143	0	0	4	22	119	554	1072	998	334	38	1	0
Wind	WSAvg	3.3	3.9	3.9	4.0	4.1	3.3	2.8	2.4	2.4	2.5	3.0	3.5	3.7
Precipitation	PrecAvg	1158	62	70	102	109	132	124	85	90	105	88	105	86
	PrecMax	1874	142	188	301	248	327	249	241	252	288	263	316	226
	PrecMin	679	7	21	14	14	36	27	24	0	10	4	9	7
	PrecStd	254	41	44	63	60	60	57	46	52	68	70	72	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	23.1	26.8	29.0	31.7	34.8	37.4	37.5	34.8	29.7	25.3	20.9
		MCWB	13.6	14.3	16.0	18.3	21.2	23.4	22.8	22.8	22.2	21.0	17.1	15.3
	2%	DB	18.3	19.9	23.8	26.9	29.4	32.7	35.1	35.1	32.2	27.7	22.4	18.0
		MCWB	13.1	13.4	15.4	17.4	21.1	23.2	23.6	23.4	22.0	19.4	15.1	13.6
	5%	DB	16.0	17.6	21.4	25.0	27.9	31.3	33.2	33.5	30.4	25.7	20.4	15.9
		MCWB	11.1	12.3	13.9	16.5	20.5	22.9	23.9	23.4	21.7	17.9	14.8	11.7
	10%	DB	13.0	15.3	19.1	22.9	26.3	30.0	32.1	31.9	28.5	23.3	18.0	13.0
		MCWB	8.9	10.4	13.0	15.7	19.6	22.5	23.7	23.1	20.6	17.2	13.2	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	16.8	18.0	20.3	23.6	25.6	26.2	26.2	24.3	22.5	18.8	16.9
		MCDB	19.0	20.6	24.3	26.1	28.9	31.6	32.3	32.1	30.4	27.3	22.7	19.4
	2%	WB	14.4	15.4	16.6	19.1	22.4	24.6	25.5	25.2	23.4	21.2	17.2	14.9
		MCDB	17.2	18.5	22.0	24.2	27.6	30.6	31.5	31.3	29.2	25.3	20.5	17.5
	5%	WB	12.0	13.3	15.3	18.2	21.5	23.9	24.9	24.5	22.7	19.8	15.7	12.5
		MCDB	14.6	16.6	19.7	22.8	26.3	29.5	30.8	30.7	28.1	23.4	19.1	14.9
	10%	WB	9.1	10.7	13.9	17.1	20.6	23.2	24.3	23.8	21.9	18.5	14.1	10.0
		MCDB	12.5	14.5	18.3	21.4	24.9	28.4	30.0	29.8	26.7	22.1	17.1	12.3
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.5	11.4	11.6	10.6	10.7	10.8	11.2	11.8	11.6	10.8	9.7
		MCDBR	13.3	13.9	14.1	13.7	11.7	12.0	12.3	13.0	13.5	13.4	13.3	12.5
		MCWBR	8.9	9.3	7.7	7.0	5.4	4.9	4.1	4.3	5.1	6.5	8.1	8.5
	5% WB	MCDBR	11.5	12.6	11.9	11.3	10.4	10.7	10.7	11.1	11.3	11.2	11.2	11.3
		MCWBR	9.1	9.7	7.9	6.9	5.3	5.1	4.5	4.4	4.9	6.3	8.3	9.1
Clear-Sky Solar Irradiance	taub	0.301	0.304	0.338	0.378	0.409	0.425	0.453	0.451	0.385	0.342	0.313	0.292	
	taud	2.506	2.497	2.434	2.324	2.290	2.265	2.244	2.221	2.390	2.494	2.527	2.545	
	Ebn at Noon	901	940	931	904	876	858	831	825	869	884	878	890	
	Edh at Noon	81	91	107	126	132	136	138	138	111	91	79	73	
All-Sky Solar Radiation	RadAvg	2.45	2.98	4.05	5.12	5.63	6.42	6.23	5.71	4.89	3.68	2.63	2.09	
	RadStd	0.21	0.38	0.25	0.30	0.48	0.50	0.37	0.40	0.39	0.46	0.29	0.26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page